

Code-Layout, Readability and Reusability

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper indentation maintained throughout the project
2	Proper File Structure	Files and folders are logically organized	yes	Separate folders for data, models, templates and static files
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Easy to understand variable names are used
4	Function / Method Names	Functions are descriptively named	yes	Functions like train_model() and predict_crop() improve readability
5	Code Comments	Inline and block comments explain logic	yes	Important sections are commented for better understanding
6	Modular Design	Code is split into reusable functions/modules	yes	Training, prediction and frontend are separated into modules
7	No Redundant Code	Duplicate or unused code is removed	yes	Optimized code with minimal redundancy
8	Error Handling	Exceptions and errors are handled gracefully	yes	Flask validation and Python exception handling are implemented

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Machine Learning Model	Python, Scikit-learn	Crop Prediction	High

2	Flask Backend API	Python, Flask	Web Application	High
3	HTML User Interface	HTML5	Frontend	High
	CSS Styling	CSS3	User Interface	High
5	JavaScript API Calls	JavaScript	Prediction Module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules
Readability	5	Easy-to-read code with meaningful names and comments
Reusability	5	Components can be reused in future smart farming projects
Documentation / Comments	5	Proper comments and documentation provided
Overall Score	5/5	Excellent code quality with good maintainability